

Shivam Medam

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Professional Summary

- Data Engineer with 5+ years of experience building and optimizing scalable batch and streaming data pipelines, ETL/ELT workflows, and analytics-ready datasets across healthcare operations, financial services, and supply chain environments.
- Strong hands-on expertise in Python, SQL, PySpark, Spark, Airflow, AWS Glue, dbt, Snowflake, and BigQuery, with a focus on reliable data processing, workflow orchestration, and warehouse performance tuning at scale.
- Experienced in translating complex business requirements into governed data pipelines and trusted transformation layers that support reporting, operational decision-making, compliance, and downstream analytics.
- Built production-grade data workflows with schema validation, data quality checks, monitoring, alerting, lineage, release controls, and automation that reduced manual effort and improved reporting reliability.
- Skilled in performance optimization, pipeline stability, and production support, with a track record of improving throughput, reducing latency and cost, and strengthening SLA adherence in fast-paced environments.
- Effective in cross-functional settings, partnering with engineering, analytics, finance, risk, and operations stakeholders to deliver scalable, audit-ready data solutions in regulated domains.

Impact Highlights: Delivered measurable impact across production data platforms, including 35% faster batch ETL, 25% lower storage cost, dashboard response time improved from 45 seconds to under 8 seconds, and reporting automation reduced from 6 hours to under 30 minutes.

Experience

Data Engineer, CVS Health | Seattle, WA

June 2025 – Present

- Rebuilt PySpark and AWS Glue pipelines for fulfillment-status and order-movement feeds that were landing out of sync across daily cycles, improving batch throughput by 35% and reducing downstream reporting gaps.
- Built and managed Apache Airflow workflows for replenishment, fulfillment, and customer-order datasets, including retries, backfills, dependency validation, and alerting that kept downstream operations reporting stable during peak-demand windows.
- Ingested Kafka and Kafka Connect order events and clickstream signals into BigQuery and Snowflake, giving business and operations teams earlier visibility into demand shifts and fulfillment bottlenecks.
- Designed centralized Snowflake and BigQuery transformation layers for fulfillment, replenishment, order, and service-exception data, creating a consistent source of truth for downstream operational reporting and analysis.
- Built and maintained dbt models and tests for curated transformation layers in Snowflake and BigQuery, improving consistency of reporting definitions and reducing rework during source changes.
- Developed an AI-assisted workflow across Snowflake, BigQuery, and internal APIs to generate refill-gap and service-exception datasets, reducing weekly analysis time from 6 hours to under 30 minutes.
- Reworked S3 and GCS storage layouts with clearer partition strategy, retention rules, and schema-version handling, lowering storage cost by 25% and improving recovery when upstream feeds changed.
- Tuned warehouse sizing, materialized views, and high-cost joins that were driving dashboard latency, improving response time from 45 seconds to under 8 seconds.
- Built Great Expectations checks for Snowflake and BigQuery pipelines using Python and Airflow, catching freshness, volume, null-rate, and schema-drift issues before they impacted operational and finance SLAs.

Data Engineer, Wells Fargo | Irving, TX

February 2024 – May 2025

- Developed distributed Scala and Spark batch pipelines to process high-volume banking datasets (customer, account, transaction activity), producing trusted inputs for risk exposure reporting, fraud monitoring, and portfolio performance analytics.
- Managed data availability, retention, and access patterns across Cassandra and Hive for structured and semi-structured banking feeds, supporting timely reporting, reconciliation workflows, and control-driven downstream consumption.
- Built reusable data models and standardized business definitions for lending trends, spend behavior, and customer segmentation, improving lineage clarity, reporting consistency, and downstream reconciliation across business and compliance teams.
- Implemented fault-tolerant Hadoop workflows with retries, checkpointing, and traceable processing steps to reduce batch failures and preserve lineage required for audit, internal controls, and regulatory review.

- Optimized Spark and Hive performance through partitioning, join strategy tuning, executor sizing, and query-plan review, reducing end-to-end runtimes by up to 30% and improving SLA adherence for daily risk, compliance, and operational reporting cycles.
- Automated validation and reconciliation checks across customer, account, and transaction data pipelines, reducing manual review effort and improving control alignment for risk, compliance, and audit reporting.
- Added release controls for Spark and Scala jobs by validating transformation logic, reconciling representative partitions, and testing high-risk schema changes in CI, reducing production regressions during peak reporting periods.
- Partnered with business, compliance, and technology stakeholders to improve data quality, governance, reconciliation, and issue remediation processes, helping align reporting datasets with SOX and internal audit expectations.

Data Engineer, CorEvitas | India

March 2020 – March 2023

- Built and maintained curated data marts and SAS datasets for registry and research reporting, improving dataset structure and access patterns and reducing retrieval time by 30% for clinical and analytics teams.
- Developed and supported 50+ operational and analytics reports using SQL, SSRS, and SSIS, delivering reliable KPI, study, and registry outputs for internal stakeholders and client-facing reporting needs.
- Designed and deployed Snowflake ETL pipelines for structured and semi-structured healthcare sources using containerized Apache Airflow on AWS ECS, reducing processing latency by 40% and stabilizing daily refresh schedules.
- Engineered streaming ingestion workflows by consuming Kafka topics and writing processed events into HBase, enabling near-real-time feed monitoring and faster operational issue detection.
- Automated repeatable data movement and infrastructure provisioning using Ansible, SNS, and AWS Lambda, reducing manual intervention and improving control over routine migration and refresh workflows.
- Implemented validation and exception-handling checks for registry feeds using Python and SQL, identifying missing fields, schema mismatches, and corrupt records before they affected downstream reporting.
- Standardized source-to-target transformations and data handling logic across registry pipelines, improving consistency, traceability, and handoff reliability for downstream analytics and reporting teams.
- Added scheduling, restart, and dependency controls to Airflow-managed workflows, improving operational stability and reducing missed refreshes across daily and weekly processing cycles.

Skills

Programming: Python, SQL, PySpark, Shell Scripting/Bash, Scala, Java

Cloud & Data Platforms: AWS (S3, EC2, EMR, Athena, Lambda, Step Functions, DMS), Azure (Data Lake, Synapse, Databricks, Data Factory), GCP (BigQuery), Snowflake, Spark

Lakehouse & Storage: Delta Lake, Apache Iceberg, Redshift, Oracle, Cassandra, MongoDB, JSON, XML, CSV

Data Engineering & ETL: Apache Airflow, AWS Glue, Kafka, Kafka Connect, AWS Lambda, Databricks, ETL Pipeline Design, Data Quality Validation, Monitoring, Performance Tuning, dbt (models, tests, docs)

Lineage, Governance & Testing: Great Expectations, OpenLineage, data catalog/lineage tools, Pytest, audit-ready data preparation

GenAI / Agentic Systems: AWS Bedrock, AI Agents (tool/function calling, orchestration, workflow automation), MCP (Model Context Protocol) for tool integration and agent interoperability

DevOps & CI/CD: Git, Docker, Jenkins

Healthcare Analytics & Compliance: EHR/EMR Data Processing, Claims Analytics, Population Health Modeling, HEDIS, HIPAA, FDA (21 CFR Part 11)

Financial Analytics & Compliance: Credit Risk Modeling, Fraud Analytics, AML/KYC, SOX & CCAR Reporting, Audit-Ready Data Preparation

Education

University of Texas at Arlington, Master's in Data Science

- **Coursework:** Database Systems, Data Warehousing & Modeling, Distributed Systems, Cloud Computing, Probability & Statistics, Machine Learning (AI).